

CraftHub Infrastructure Exchange

Research & Evidence Report From Fragmented Capacity to Demand-Driven Innovation Infrastructure

Prepared for Smart India Hackathon 2026 — Synergy Squad

1. Executive Summary

CraftHub proposes a demand-driven infrastructure network connecting students, innovators, startups, researchers, MSMEs and businesses with public, private and institutional laboratories, testing facilities, fabrication centres, machines and manufacturing capacity. The objective is not to replace existing government platforms such as I-STEM, but to connect fragmented demand and supply across infrastructure types and convert project requirements into executable physical work.

Government evidence supports access, discoverability, prototyping, testing, validation, coordination and utilization challenges. It does not justify claiming a specific percentage of India's laboratories is idle or a specific percentage lies outside I-STEM. Those figures require a dedicated cross-dataset infrastructure census.

2. What Government Evidence Establishes

NITI Aayog — Ease of Doing R&D in India (2026): limited access to TRL 3–6 prototyping, pilot-scale validation, testing/certification infrastructure, waiting times, technical support and fragmented support mechanisms.

I-STEM — Government of India: a national mechanism for discovering and accessing scientific equipment; its rationale addresses equipment being distributed across institutions and difficult for potential users to discover or access.

DST — FIST / FIST 2.0: wider access and utilization of scientific infrastructure, including use by startups, manufacturing industries and MSMEs.

DST — SATHI: shared, professionally managed infrastructure for academia, startups, industry and other users.

MSME Ministry — Common Facility Centres: distributed shared infrastructure for testing, training, production and related cluster requirements.

3. India's R&D Ecosystem Is Larger Than I-STEM

The NSTMIS Directory of R&D Institutions 2025 reports 8,252 R&D institutions in India, including 5,394 private-sector institutions (65.4%), 1,048 state-sector institutions, 1,034 higher-education institutions, 622 central-sector institutions and 154 public-sector industry institutions.

This must not be converted directly into an I-STEM coverage ratio. I-STEM and NSTMIS use different definitions and datasets. I-STEM focuses on scientific equipment/infrastructure made accessible through its platform, while NSTMIS is a broader R&D-institution directory. Therefore, a calculation such as $3,974 / 8,252$ is not a valid national coverage percentage.

The missing coverage figure should be presented as a research gap to be measured, not as an assumed statistic.

4. The Proposed Niche: Physical Innovation Workspace

CraftHub should not position itself merely as a laboratory-slot booking platform. The proposed niche is a cross-sector demand-and-execution layer.

PROJECT / IDEA

↓

TECHNICAL REQUIREMENTS

↓

STANDARDIZED PHYSICAL-WORK REQUEST

↓

CAPABILITY + LOCATION + CAPACITY MATCHING



PROVIDER / QUOTE / SCHEDULE



PHYSICAL EXECUTION



TEST / FABRICATE / BUILD



RESULT + EVIDENCE BACK TO PROJECT



NEXT ITERATION / MORE DEMAND

5. Supply–Demand Expansion Mechanism

The strategic extension beyond infrastructure-access systems is demand aggregation. A small number of fragmented requests may be individually unattractive to a provider, while aggregated recurring demand can reveal a commercially meaningful market.

Example: multiple buyers separately request small PCB jobs. CraftHub can aggregate valid requirements by capability, geography, specification, timeline and quantity. A provider sees current and recurring demand rather than isolated discovery. Persistent unmet demand can become a signal for new machines, labs, fabrication businesses or capacity expansion.

6. Demand → Capacity → Innovation Loop

Demand: students / startups / researchers / businesses / innovators



Standardized requirement



CraftHub demand engine



Existing capacity matched OR missing capacity detected



Provider opportunity / investment signal



New or expanded capacity



Physical execution



Prototype / test / product / research evidence



More projects and recurring demand



7. What We Can Claim — and What We Should Not Claim

8. Proof Plan for CraftHub

Provider supply: facilities, machines, capabilities and available capacity mapped.

Demand: standardized requests by category and geography.

Matching: percentage of valid requests successfully matched.

Access: median time from request to provider/quote.

Execution: completion, rejection, cancellation and turnaround rates.

Utilization: provider capacity before and after participation.

Economics: incremental revenue from previously available capacity.

Innovation: projects reaching prototype, testing, pilot or validated result.

Gap detection: recurring unmet requirements by geography/capability.

Capacity response: new providers, machines or services added after sustained demand signals.

9. Proposed Pilot

Start with one geographic innovation cluster. Map demand from students, startups, researchers, MSMEs and businesses against supply from universities, private labs, testing facilities, makerspaces, CNC/3D-printing providers, PCB manufacturers, CFCs and other relevant facilities.

Run a before/after comparison of search time, matching rate, turnaround time, utilization, cost and prototype completion. If sustained unmet demand appears, test whether providers respond by adding or expanding capacity.

10. Proposed Positioning

CraftHub Infrastructure Exchange

From fragmented capacity to demand-driven innovation infrastructure.

CraftHub does not need to own every laboratory, machine, factory or logistics network. It coordinates demand, capability, access, execution and evidence across participating providers.

11. References — Official Government / Institutional Sources

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12. Research Limitation & Next Dataset

A national 'I-STEM-accessible versus non-I-STEM-accessible laboratory ratio' should not be reported until the underlying populations are harmonized. The recommended next study is an India Infrastructure Census combining I-STEM, NSTMIS, DSIR, NABL, MSME CFCs, Technology Centres and additional public/private facility registries, followed by deduplication, capability classification and geographic mapping.

Defensible	Do not claim without new measurement
India has a substantial distributed R&D/infrastructure ecosystem.	X% of all Indian labs are idle.
Government sources identify access, prototyping, testing, validation and coordination barriers.	X% of labs are outside I-STEM.
I-STEM exists to improve discovery/access/utilization of scientific infrastructure.	I-STEM is ineffective.
Private R&D is a major part of India's R&D institution base.	All private labs are inaccessible.
Fragmented demand can be tested as a barrier to capacity expansion.	New labs/fabs do not emerge because customers do not exist.